

UHIP

Unified Healthcare Intelligence Platform

Port Said Governorate — Egypt

Data Warehouse — Dimensional Model Design

Version 1.0 | May 2025

1. Overview

This document presents the Data Warehouse Dimensional Model for UHIP. It covers the schema type, identified fact and dimension tables, a Bus Matrix, and full column-level table definitions for all 25 source tables mapped to their DWH counterparts.

1.1 Schema Type

Schema: Fact Constellation (Galaxy Schema)

The UHIP DWH uses a Fact Constellation schema — three fact tables sharing conformed dimensions. Each fact table alone forms a star schema; together they form a galaxy/constellation. This design suits the domain because there are three distinct business processes (clinical encounters, insurance claims, pharmacy movements) that must be analysed independently and jointly across shared dimensions such as date, hospital, and patient.

1.2 Grain Summary

Fact Table	Grain (one row per...)	Primary Source
fact_visits	Patient encounter (visit)	visits
fact_claims	Insurance claim submission	claims
fact_pharmacy	Drug inventory transaction	drug_transactions
fact_claim_items		Claim_items
Fact_referral		referral

2. Bus Matrix

The Bus Matrix shows which dimensions are shared across fact tables. ✓ = dimension applies to this fact. — = not applicable.

Dimension	fact_visits	fact_claims	fact_pharmacy	fact_claim_items	fact_referrals
dim_date	✓	✓	✓	✓	✓
dim_patient	✓	✓	—		✓
dim_hospital	✓	✓	✓		✓
dim_doctor	✓	—	—		
dim_department	✓	—	—		
dim_diagnosis	✓	—	—		
dim_procedure	✓	✓	—	✓	
dim_drug	—	✓	✓	✓	
dim_claim_status	—	✓	—		

Conformed dimensions (shared by 2+ facts): dim_date, dim_hospital, dim_patient, dim_procedure, dim_drug.

3. Fact Tables

3.1 fact_visits

Central fact table. One row per completed patient encounter. Grain: one visit. All clinical activity references a visit.

Source tables: visits, visit_procedures, medical_records, prescriptions, prescription_items, patient_feedback

Column	Type	Key	Description
visit_id	VARCHAR(12)	PK	Unique visit identifier
date_key	INT	FK	→ dim_date (visit_date)
patient_key	INT	FK	→ dim_patient
hospital_key	INT	FK	→ dim_hospital
doctor_key	INT	FK	→ dim_doctor
department_key	INT	FK	→ dim_department
diagnosis_key	INT	FK	→ dim_diagnosis
visit_type	VARCHAR(15)		Emergency / Inpatient / Outpatient / Follow-up / Routine Check
total_cost	DECIMAL(10,2)		Total billed amount (EGP)
waiting_time	INTEGER		Minutes waited before seen
procedure_count	INTEGER		Number of procedures performed
prescription_flag	TINYINT		1 if prescription generated
feedback_rating	DECIMAL(3,1)		Patient satisfaction rating (1–5, NULL if none)
visit_status	VARCHAR(10)		Completed / No Show / Cancelled

3.2 fact_claims

Insurance claims fact. One row per claim. Grain: one insurance claim submission. Linked 1:1 to a visit.

Source tables: claims, claim_items, claim_approvals

Column	Type	Key	Description
claim_id	VARCHAR(12)	PK	Unique claim identifier
visit_id	VARCHAR(12)	FK	→ fact_visits (claim linked to visit)
date_key	INT	FK	→ dim_date (claim_date)
patient_key	INT	FK	→ dim_patient
hospital_key	INT	FK	→ dim_hospital
procedure_key	INT	FK	→ dim_procedure (primary billed procedure)
drug_key	INT	FK	→ dim_drug (primary billed drug, nullable)
claim_status_key	INT	FK	→ dim_claim_status
claim_amount	DECIMAL(10,2)		Total claimed amount (EGP)
approved_amount	DECIMAL(10,2)		Amount approved by insurer (NULL if pending)
approval_gap	DECIMAL(10,2)		claim_amount – approved_amount
days_to_decision	INTEGER		Days between claim_date and approval_date

Column	Type	Key	Description
item_count	INTEGER		Total line items in claim

3.3 fact_pharmacy

Pharmacy movements fact. One row per drug transaction. Grain: one drug movement event at a hospital.

Source tables: drug_transactions, drug_inventory, suppliers

Column	Type	Key	Description
transaction_id	VARCHAR(9)	PK	Unique transaction identifier
date_key	INT	FK	→ dim_date (transaction_date)
drug_key	INT	FK	→ dim_drug
hospital_key	INT	FK	→ dim_hospital
supplier_key	INT	FK	→ dim_supplier (Purchase transactions only)
transaction_type	VARCHAR(12)		Purchase / Dispensing / Wastage / Return / Adjustment
quantity	INTEGER		Units moved
unit_price	DECIMAL(8,2)		Price per unit at time of transaction (EGP)
total_value	DECIMAL(10,2)		quantity × unit_price
stockout_flag	TINYINT		1 if stock fell to 0 after transaction
near_expiry_flag	TINYINT		1 if batch expiry within 60 days
no_prescription_flag	TINYINT		1 if Dispensing with no matching prescription (fraud signal)

4. Dimension Tables

4.1 dim_date

Date dimension. One row per calendar day. Covers full study period 2024-05-09 to 2025-05-09. Derived — no source table.

Column	Type	Key	Description
date_key	INT	PK	Surrogate key (YYYYMMDD format)
full_date	DATE		Calendar date
year	SMALLINT		Calendar year
quarter	TINYINT		Quarter (1–4)
month	TINYINT		Month number (1–12)
month_name	VARCHAR(10)		January ... December
week	TINYINT		ISO week number
day_of_week	TINYINT		1=Monday ... 7=Sunday
is_weekend	TINYINT		1 if Saturday or Sunday
season	VARCHAR(10)		Winter / Spring / Summer / Autumn

4.2 dim_patient

Patient dimension. Source: patients table.

Column	Type	Key	Description
patient_key	INT	PK	Surrogate key
patient_id	VARCHAR(12)		Natural key (PAT + 6 digits)
national_id	VARCHAR(14)		Egyptian national ID
first_name	VARCHAR(50)		Patient first name
last_name	VARCHAR(50)		Patient last name
gender	VARCHAR(6)		Male / Female
birth_date	DATE		Date of birth
age_group	VARCHAR(15)		Derived: Child / Adult / Senior
city	VARCHAR(20)		District in Port Said
governorate	VARCHAR(20)		Always Port Said
blood_type	VARCHAR(3)		ABO/Rh blood group

4.3 dim_hospital

Hospital dimension. Source: hospitals table.

Column	Type	Key	Description
hospital_key	INT	PK	Surrogate key

Column	Type	Key	Description
hospital_id	VARCHAR(4)		Natural key (H001–H008)
hospital_name	VARCHAR(80)		Official facility name
hospital_type	VARCHAR(15)		Government / Private / Specialized / Teaching
district	VARCHAR(25)		Port Said district
address	VARCHAR(100)		Street address
total_beds	INTEGER		Licensed bed capacity
icu_capacity	INTEGER		Dedicated ICU bed count
latitude	DECIMAL(9,6)		Geographic latitude
longitude	DECIMAL(9,6)		Geographic longitude

4.4 dim_doctor

Doctor dimension. Source: doctors table.

Column	Type	Key	Description
doctor_key	INT	PK	Surrogate key
doctor_id	VARCHAR(7)		Natural key (DOC + 4 digits)
first_name	VARCHAR(50)		Doctor first name
last_name	VARCHAR(50)		Doctor last name
specialty	VARCHAR(40)		Medical specialty
years_experience	INTEGER		Years in medical practice
employment_status	VARCHAR(10)		Active / On Leave
hospital_key	INT	FK	→ dim_hospital
department_key	INT	FK	→ dim_department

4.5 dim_department

Department dimension. Source: departments table.

Column	Type	Key	Description
department_key	INT	PK	Surrogate key
department_id	VARCHAR(8)		Natural key (DEPT + 4 digits)
department_name	VARCHAR(40)		E.g. Emergency, ICU, Cardiology
floor_number	VARCHAR(2)		Floor location (1 / 2 / 3)
hospital_key	INT	FK	→ dim_hospital

4.6 dim_diagnosis

Diagnosis dimension. Source: diagnoses table. 120 ICD-style codes.

Column	Type	Key	Description
diagnosis_key	INT	PK	Surrogate key
diagnosis_code	VARCHAR(4)		Natural key (D001–D120)
diagnosis_name	VARCHAR(100)		Full clinical diagnosis name
diagnosis_category	VARCHAR(30)		Disease system (Cardiovascular, Respiratory, etc.)
severity_level	VARCHAR(10)		Mild / Moderate / Severe / Critical / Chronic

4.7 dim_procedure

Procedure dimension. Source: procedures table. 201 procedures from Egyptian healthcare services list.

Column	Type	Key	Description
procedure_key	INT	PK	Surrogate key
procedure_code	VARCHAR(4)		Natural key (P001–P201)
procedure_name	VARCHAR(120)		Full procedure name
procedure_category	VARCHAR(30)		Laboratory / Radiology / Surgery / etc.
expected_cost	DECIMAL(10,2)		Standard reference price (EGP)
complexity_score	INTEGER		1 (simple lab) to 5 (ICU/major surgery)

4.8 dim_drug

Drug dimension. Source: drugs table. 175 unique drugs, 21 therapeutic categories.

Column	Type	Key	Description
drug_key	INT	PK	Surrogate key
drug_id	VARCHAR(5)		Natural key (DR001–DR175)
drug_name	VARCHAR(80)		Brand/trade name with strength and formulation
generic_name	VARCHAR(60)		International non-proprietary name (INN)
manufacturer	VARCHAR(50)		Egyptian manufacturer or local subsidiary
drug_category	VARCHAR(30)		Therapeutic class (Antibiotic / Antidiabetic / etc.)
unit_price	DECIMAL(8,2)		Price per unit (EGP), EGP 5 – EGP 1,146

4.10 dim_claim_status

Claim status dimension. Derived from claims.claim_status and claim_approvals.approval_status.

Column	Type	Key	Description
claim_status_key	INT	PK	Surrogate key
claim_status	VARCHAR(20)		Approved / Partially Approved / Rejected / Pending Review
rejection_reason	VARCHAR(80)		Populated for Rejected claims; NULL otherwise
reviewed_by	VARCHAR(50)		Insurer staff reviewer name

5. Source Table → DWH Mapping

All 25 source tables from the UHIP logical design mapped to their DWH destination.

Source Table	DWH Destination	Role
patients	dim_patient	Patient dimension
visits	fact_visits	Primary fact table
diagnoses	dim_diagnosis	Diagnosis dimension
medical_records	fact_visits	Merged: notes & follow-up flag into fact_visits
visit_procedures	fact_visits	Aggregated: procedure_count measure
prescriptions	fact_visits	Merged: prescription_flag into fact_visits
prescription_items	fact_visits + fact_claims	Quantity measures; linked to drug_key
hospitals	dim_hospital	Hospital dimension
departments	dim_department	Department dimension
doctors	dim_doctor	Doctor dimension
doctor_schedules	fact_visits (junk)	Shift type joined at load for overtime analysis
beds	dim_hospital	Bed counts aggregated into dim_hospital
icu_status	Separate ICU snapshot fact (optional)	Time-series ICU occupancy
referrals	Bridge / degenerate dim	Referral reason as degenerate dimension on fact_visits
drugs	dim_drug	Drug dimension
drug_inventory	fact_pharmacy	Stock snapshot measures
drug_transactions	fact_pharmacy	Primary source for pharmacy fact
suppliers	dim_supplier	Supplier dimension
claims	fact_claims	Primary source for claims fact
claim_items	fact_claims	Line-item aggregation; drug_key / procedure_key
claim_approvals	fact_claims + dim_claim_status	Approval outcome and rejection reason
procedures	dim_procedure	Procedure dimension
patient_feedback	fact_visits	feedback_rating measure aggregated into fact_visits

Source Table	DWH Destination	Role
diagnoses (ref)	dim_diagnosis	Already listed above
procedures (ref)	dim_procedure	Already listed above

6. Key Analytical Use Cases

The following analytical questions are directly answerable from the dimensional model:

Use Case	Facts Used	Dimensions Joined
Monthly visit volume by hospital type	fact_visits	dim_date, dim_hospital
Average waiting time govt vs private	fact_visits	dim_hospital, dim_visit_type
Top diagnoses by severity and season	fact_visits	dim_diagnosis, dim_date
Doctor workload and overtime rate	fact_visits	dim_doctor, dim_date
ICU occupancy trend (winter peak)	fact_visits	dim_hospital, dim_date
Insurance approval rate by hospital	fact_claims	dim_hospital, dim_claim_status
Claim rejection reasons distribution	fact_claims	dim_claim_status
Overbilling detection (inflated procedures)	fact_claims + fact_visits	dim_procedure, dim_hospital
Drug dispensing without prescription (fraud)	fact_pharmacy	dim_drug, dim_hospital, dim_date
Near-stockout and near-expiry alerts	fact_pharmacy	dim_drug, dim_hospital
Patient satisfaction vs waiting time	fact_visits	dim_hospital, dim_doctor
Referral patterns between hospitals	fact_visits	dim_hospital (from/to)